

White noise process:

$\{X_t\}$ is a sequence of uncorrelated r.v.s such that

$$\mathbb{E}(X_t) = \mu, \quad V(X_t) = \sigma^2, \quad \forall t$$

$$S_t = \text{Cov}(X_t, X_{t+z}) = \begin{cases} \sigma^2 & \text{if } z=0 \\ 0 & \text{otherwise} \end{cases}$$

Note: white noise $\Rightarrow$ iid. Consider if $X_i = \begin{cases} \mathcal{N}(0,1) & \text{if } i \text{ is even} \\ \text{Uniform}(-\sqrt{3}, \sqrt{3}) & \text{otherwise} \end{cases}$

and $\{X_t\}$ are independent sequence.

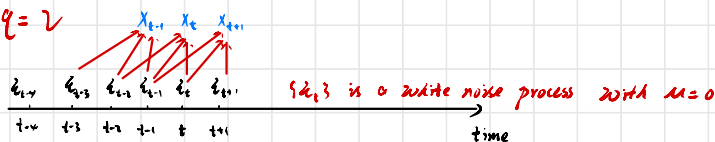
When i is even, note that $\mathbb{E}(X) = 0$

$$\begin{aligned} \mathbb{E}(X^2) &= \int_a^b \frac{1}{a-b} x^2 dx \\ &= \frac{1}{a-b} \frac{x^3}{3} \Big|_{x=a}^b \\ &= \frac{1}{a-b} \left(\frac{b^3 - a^3}{3} \right) \\ &= \frac{1}{2\sqrt{3}} \frac{3\sqrt{3} + 3\sqrt{3}}{3} \\ &= 1 \end{aligned} \quad \Rightarrow \quad V(X) = 1$$

So the process is white noise, however not iid

Moving average process MA(q)

example: $q=2$



$$X_t = \mu - \sum_{i=0}^q \theta_i \varepsilon_{t-i}$$

X_t is a stationary process

$$\begin{aligned} \text{Cov}(X_t, X_{t+2}) &= \bar{E}(X_t X_{t+2}) - \bar{E}(X_t) \bar{E}(X_{t+2}) \\ &= \bar{E} \left(\left(\mu - \sum_{i=0}^q \theta_i \varepsilon_{t-i} \right) \left(\mu - \sum_{i=0}^q \theta_i \varepsilon_{t+2-i} \right) \right) - \mu^2 \\ &= \bar{E} \left(\mu^2 - \mu \sum_{i=0}^q \theta_i \varepsilon_{t+2-i} - \mu \sum_{i=0}^q \theta_i \varepsilon_{t-i} + \left(\sum_{i=0}^q \theta_i \varepsilon_{t-i} \right) \left(\sum_{i=0}^q \theta_i \varepsilon_{t+2-i} \right) \right) - \mu^2 \\ &= \bar{E} \left(\left(\sum_{i=0}^q \theta_i \varepsilon_{t-i} \right) \left(\sum_{i=0}^q \theta_i \varepsilon_{t+2-i} \right) \right) \\ &= \begin{cases} \sum_{i=0}^q \sum_{j=0}^q \theta_i \theta_{i+2} & \text{if } 2 \leq q \\ 0 & \text{if } 2 > q \end{cases} \end{aligned}$$

depends on q only

Estimates the parameters:

$$\theta_0 \hat{\varepsilon}_1 = (X_1 - \mu)$$

assume $\theta_{-1} = 0$

$$\theta_0 \hat{\varepsilon}_2 = (X_2 - \mu) - \theta_1 \hat{\varepsilon}_1$$

:

$$\theta_0 \hat{\varepsilon}_t = (X_t - \mu) - \sum_{j=1}^q \theta_j \hat{\varepsilon}_{t-j}$$

then we solve $\underset{\mu, \theta}{\text{argmin}} \sum_{t=1}^n \hat{\varepsilon}_t^2$

Making predictions:

$$\begin{aligned} \bar{E}(X_{t+n} | X_1, \dots, X_t) &= \bar{E} \left(\mu - \sum_{i=1}^q \theta_i \varepsilon_{t+n-i} | X_1, \dots, X_t \right) \\ &= \mu - \sum_{i=1}^q \theta_i \bar{E}(\varepsilon_{t+n-i} | X_1, \dots, X_t) - \theta_0 \bar{E}(\varepsilon_{t+n}) \\ &= \mu - \sum_{i=1}^q \theta_i \bar{E}(\varepsilon_{t+n-i} | X_1, \dots, X_t) \end{aligned}$$

So we need to estimate ε_{t+n-i} given all observations

How do we predict $z_i | x_1, \dots, x_k$?

Forward Recursion: given the observed data sequence $X_{1:t}$,

we predict $\hat{x}_t = (x_t - \hat{x}_t) / \hat{\sigma}_t$

$$\Rightarrow \hat{\theta}_t \hat{z}_t = (\mu - \sum_{i=0}^p \theta_i \hat{z}_{t-i} - \hat{\mu} + \sum_{i=1}^p \hat{\theta}_i \hat{z}_{t-i})$$

$$\Rightarrow \hat{\theta}_0 \hat{z}_t - \theta_0 z_t = (\mu - \hat{\mu}) + \sum_{i=1}^p (\theta_i z_{t-i} - \hat{\theta}_i \hat{z}_{t-i})$$

Let $\hat{\theta}_i \rightarrow \theta_i, \hat{\mu} \rightarrow \mu$

$$\Rightarrow \theta_0(z_t - \hat{z}_t) = \sum_{i=1}^p \theta_i (z_{t-1} - \hat{z}_{t-1})$$

$$\Rightarrow \theta_0 \cdot e_t = \sum_{i=1}^p \theta_i \cdot e_{t-i}$$

if $p=1 \Rightarrow \theta_0 e_t = \theta_1 e_{t-1}$

$$\Rightarrow l_t = \frac{\theta_1}{\theta_0} l_{t-1}$$

$$\Rightarrow e_t = \left(\frac{\theta_1}{\theta_0} \right)^t e_0$$

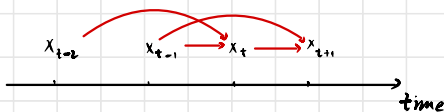
✓) when $\frac{\sigma_1}{\sigma_0} < 1$, Converge to 0, So the estimation gets better and better when t increases.

however, when $\frac{\theta_1}{\theta_0} > 1$, the estimate error accumulates $\rightarrow \infty$

Autoregressive process $AR(q)$

$$X_t = \phi_1 X_{t-1} + \dots + \phi_q X_{t-q} + \varepsilon_t$$

when $q=2$



Note: not all AR process are stationary

Consider $AR(1)$, where $X_t = \phi X_{t-1} + \varepsilon_t$

$$\begin{aligned} \Rightarrow X_t &= \phi(\phi X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \\ &= \phi^2 X_{t-2} + \phi \varepsilon_{t-1} + \varepsilon_t \\ &= \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \dots \\ &= \sum_{k=0}^{t-1} \phi^k \varepsilon_{t-k} + \underbrace{\phi^t X_0}_{\rightarrow \text{wlog let } X_0 = 0} \\ &= \sum_{k=0}^{t-1} \phi^k \varepsilon_{t-k} \end{aligned}$$

$$\text{So } E(X_t) = 0$$

$$\begin{aligned} V(X_t) &= E(X_t^2) = E\left(\sum_{k=0}^{t-1} \phi^k \varepsilon_{t-k}\right)^2 \\ &= \sum_{k=0}^{t-1} \phi^{2k} \sigma^2 \\ &= \sigma^2 \frac{(1 - \phi^{2t})}{1 - \phi^2} \end{aligned}$$

So for $V(X_t) < \infty$, we must have $|\phi| < 1$

$$\Rightarrow V(X_t) \approx \frac{\sigma^2}{1 - \phi^2} \text{ when } t \text{ is large enough}$$

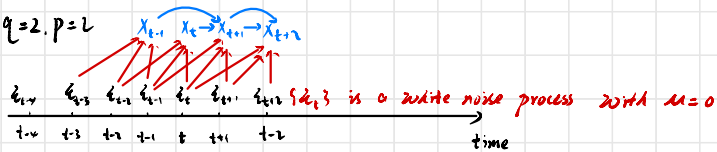
if $|\phi| < 1$ (stationary), the influence of past ε decays exponentially

if $|\phi| = 1$ (non-stationary), random walk, the effect of ε stays

if $|\phi| > 1$ (non-stationary), the effect of past ε increases exponentially

ARMA(q, p)

Example: $q=2, p=2$



$$X_t = \underbrace{\sum_{k=1}^q \phi_k X_{t-k}}_{\text{AR}(q)} + z_t - \underbrace{\sum_{k=1}^p \theta_k z_{t-k}}_{\text{MA}(p)}$$

ARIMA(q, d, p)

Define the difference operator

$$\Delta = (1 - B)$$

where $BX_t = X_{t-1}$ is the backward shift operator

Then we can rewrite ARMA(p, q) as

$$X_t - \sum_{k=1}^q \phi_k B^k X_t = z_t - \sum_{k=1}^p \theta_k B^k z_t$$

$$\underbrace{(1 - \sum_{k=1}^q \phi_k B^k)}_{\Phi(B)} X_t = \underbrace{(1 - \sum_{k=1}^p \theta_k B^k)}_{\Theta(B)} z_t$$

$$\text{ARMA: } \Phi(B) X_t = \Theta(B) z_t$$

$$\text{ARIMA: } \Phi(B) (1-B)^d X_t = \Theta(B) z_t$$

$$\text{or } \Phi(B) \Delta^d X_t = \Theta(B) z_t$$